

Green Pool Recovery: The First-Time Pool Owner's Guide to Clear Water Fast

You bought a house with a pool. The water is green. You have no idea what to do. This guide fixes that.

Page 1: The Green Water Moment

You moved into your house. Congratulations on the pool. Either the previous owner left it green, or you opened it in spring and found that nature had taken over. Either way, you're standing in your backyard right now looking at something that resembles pond water more than a swimming pool.

You called a pool company. They quoted you \$350 for a "startup service." You searched online and found advice that contradicts itself: some people say bleach, some say shock, some say you need special expensive chemicals, some say you need a professional, some say it's easy.

Here's the honest truth: green water is almost always algae. Algae is treatable. The SLAM method — Shock, Lapse, Algae, Maintain — is the most reliable way to clear a green pool. It takes 3–7 days. It costs \$50–\$150 in chemicals if you do it yourself. This guide will walk you through every step.

Page 2: Understanding What's Actually in Your Pool Water

Before you can fix the water, you need to know what you're measuring. Pool chemistry isn't complicated, but it is precise — you need to understand five key numbers.

pH (7.2–7.6): This measures how acidic or basic your water is. Below 7.2 = corrosive (etching your pool surfaces, burning swimmer eyes). Above 7.6 = scale-forming (cloudy water, chlorine becomes less effective). This is the most frequently mis-managed number. Check it 2–3 times per week.

Free Chlorine (FC): The active chlorine available to kill bacteria and algae. Without it, your pool becomes a breeding ground. During the SLAM process, you'll drive this number up significantly. In normal maintenance, keep it between 2–4 ppm for most pools.

Combined Chlorine (CC): Chlorine that's already been used up reacting with contaminants. If CC is above 0.5 ppm, it's time to shock. High CC causes "chloramine" smell — the strong smell most people associate with indoor pools.

Total Alkalinity (TA) (80–120 ppm): This is the buffer that keeps your pH stable. Low TA = pH bounces around wildly. High TA = pH rises and won't come down. Always balance TA before adjusting pH.

CYA (Stabilizer/Cyanuric Acid) (30–50 ppm): CYA protects chlorine from being destroyed by sunlight. Without it, chlorine burns off in hours. Too much CYA (above 80 ppm) "locks out" chlorine — you add it but it doesn't work. If your CYA is too high, the only fix is partially draining and refilling the pool.

Calcium Hardness (200–400 ppm): How much calcium is dissolved in the water. Too low = corrosive to plaster. Too high = cloudy water and scale. Most tap water is fine for calcium; adjust only if you're filling with very soft or very hard water.

Page 3: The SLAM Method — Your Green Pool Recovery Protocol

SLAM stands for **Shock, Lapse, Algae, Maintain**. It was developed by pool industry professionals and is the most reliable method for clearing green water. Every step matters. Don't skip around.

Step 1 — Test your water first. Before adding any chemicals, use a reliable test kit. Not test strips — they degrade with temperature and age and give inaccurate readings at the extremes that matter most. Use a liquid test kit: the Taylor K-2006 or K-2006C is the industry standard for homeowners. It gives you accurate readings for FC, CC, pH, TA, and CYA. Cost: \$30–\$45.

Step 2 — Balance Total Alkalinity first. Add sodium bicarbonate (baking soda) to raise TA to 80–100 ppm. Use the dosing chart that comes with your Taylor kit or the poolcalculator website. TA affects pH; get TA right before you chase pH numbers.

Step 3 — Adjust pH to 7.2–7.6. Use muriatic acid to lower pH (add acid to water, never add water to acid). Use soda ash (sodium carbonate) to raise pH. Small adjustments — retest after each addition and wait 30 minutes.

Step 4 — Check CYA. If you don't know your CYA level, test it. If it's above 80 ppm, stop — you need to drain 30–50% of the pool and refill before proceeding. High CYA makes chlorine useless, which is why your pool likely went green in the first place.

Step 5 — SLAM shock. With TA and pH balanced, add liquid chlorine (12% sodium hypochlorite — unscented household bleach is typically 6–8%; pool-grade liquid chlorine is 10–12%). Calculate the dose using the Taylor test kit's SLAM chart based on your CYA level. For a typical green pool with CYA at 30–50 ppm, you're targeting Free Chlorine levels of 16–24 ppm during the SLAM. Yes, that's high. Algae survives normal chlorine levels — you need shock-level chlorine to kill it.

Step 6 — Run the pump 24 hours a day. Circulation is required. The chlorine needs to move through the water. The filter needs to capture dead algae. Set your valve to FILTER (not recirculate — that bypasses the filter). Run the pump continuously until the water is clear.

Step 7 — Brush the pool. Algae clings to surfaces with a biofilm. Vacuuming alone won't remove it. Use a stiff-bristle pool brush on the walls, floor, and steps — especially corners, returns, and ladder areas. Brush once or twice daily during the SLAM.

Step 8 — Wait and retest. Check FC levels daily. The SLAM process is complete when: FC holds overnight (less than 1 ppm loss from evening to morning), CC is 0.5 ppm or below, and the water is clear (you can see the drain or main drain fitting from the surface). This typically takes 3–7 days for moderate algae. Severe algae can take 2 weeks.

Page 4: The Chemical Shopping List — What to Buy, What to Skip

You don't need most of what the pool store recommends. Here's exactly what to buy.

Liquid chlorine (10–12% sodium hypochlorite): Your primary shock product. Buy it in 1-gallon jugs from a hardware store or pool supply. Store in a cool, dark place. shelf life is 1–2 months; buy only what you'll use in a season. Cost: \$3–\$6 per gallon. **Do not** buy scented bleach — the additives are unnecessary.

Muriatic acid (31.45% hydrochloric acid): For lowering pH. Pool-grade muriatic acid is available at hardware stores and pool suppliers. Cost: \$10–\$20 per gallon. Handle with gloves and eye protection; add acid to water, slowly, with the pump running.

Sodium bicarbonate (baking soda): For raising Total Alkalinity. Food-grade is fine. 1.5 pounds raises TA by 10 ppm per 10,000 gallons of pool water. Cost: \$0.60/lb at grocery stores.

Sodium carbonate (soda ash): For raising pH. Not the same as baking soda. Cost: \$10–\$15 for a 5-pound bag at pool suppliers.

Calcium hypochlorite (cal-hypo) shock — optional: Faster dissolving than liquid chlorine, doesn't add CYA. Use only if your CYA is already low and you want a faster SLAM. Do not mix with acid or other chemicals — dangerous reaction.

CYA test reagent: If using the Taylor K-2006C, it includes CYA reagent. If not, buy CYA reagent separately — you need this number.

What to skip: Pool "clarifier" products (usually unnecessary if filtration is working); algaecide with copper (can stain pools and interfere with chlorine); expensive branded shock products (same active ingredients for 5x the price); cyanuric acid stabilizer unless your CYA test reads below 30 ppm and you need to raise it.

Page 5: Your Filter — Running It Right

The filter is doing physical work that chlorine can't do alone. If your filter isn't working correctly, adding chemicals is like filling a bucket with a hole in it.

Sand filters: The most common residential type. Works by passing water through a bed of sand that traps debris. When the pressure gauge reads 8–10 psi above the "clean" baseline (which you should record when the filter is freshly backwashed), it's time to backwash. Backwashing reverses flow through the filter and flushes the trapped debris into a drain. Run until the sight glass runs clear, usually 2–3 minutes. After backwashing, you must re-check your pressure gauge and record the new baseline.

Cartridge filters: Common on smaller pools and above-ground pools. The cartridge element traps debris. To clean: remove the cartridge, hose it off with a garden hose (high-pressure nozzle on the hose works better than just letting it sit). For deep cleaning, soak in a TSP (trisodium phosphate) solution for 12–24 hours, then rinse. Replace cartridges every 1–3 years depending on usage and water quality. Never run the pump with a removed cartridge — unfiltered water returns to the pool full of debris.

DE (diatomaceous earth) filters: Highest filtration 精度 but most maintenance. After backwashing, you must add fresh DE powder (1–2 pounds per square foot of filter area) through the skimmer to recharge the grid. Messy but effective. Follow manufacturer instructions for your specific filter model.

The recirculate mistake: Many pool systems have a "recirculate" setting on the multiport valve. This setting bypasses the filter entirely. If someone switched it to recirculate during a SLAM, you're adding chlorine to water that isn't being filtered. The

dead algae stays in the water. Check your valve position every time you add chemicals. It should be set to FILTER.

Page 6: Why Your Pool Went Green in the First Place

Understanding the cause helps you prevent a recurrence. Green water (algae) grows when two conditions overlap: insufficient chlorine and favorable conditions for biological growth.

Low or zero chlorine: The most common cause. If FC was allowed to drop to zero — even for a day or two — algae took hold. This happens when pools are not tested regularly, when pools are "winterized" without proper chemical balance, or when CYA was so high that chlorine was locked out and couldn't do its job.

High CYA blocking chlorine: If CYA is above 80 ppm, no matter how much chlorine you add, the chlorine is "protected" by the CYA molecules and can't kill algae. This is the most misdiagnosed cause of persistent green water. If your pool stays green despite heavy chlorination, check CYA first.

Poor circulation: Areas of the pool that don't receive adequate water flow become algae havens. This includes behind return jet fittings, under ladder grips, in shaded corners, and behind pool steps. If you have "green spots" rather than uniform green water, circulation is likely the issue. Adding a directional return jet can fix dead zones.

Improper winterization: Pools left uncovered over winter without proper closing chemicals develop algae blooms early in the spring. The water heats up under a winter cover, CYA and chlorine levels drop to zero, and algae grows explosively. A proper pool closing includes shocking the water, adding a winter algaecide, and lowering the water level below the skimmer.

Heavy organic load: A pool near trees receives constant leaf and pollen input. Each leaf that falls in and isn't removed consumes chlorine as it decomposes. Heavy bather load (many swimmers, sweating, sunscreen, urinated pool water) does the same thing. Large organic loads require more frequent shocking.

Page 7: Weekly Maintenance Routine — The 30-Minute Schedule

You can keep your pool clear all season with 30 minutes of attention per week. Here's the system.

Every 2–3 days (10 minutes): - Test FC and pH using your liquid test kit. Record results. - Adjust FC if below 2 ppm (add liquid chlorine); adjust pH if outside 7.2–7.6 range. - If FC drops to zero between tests, something is wrong — check CYA and for obvious contamination (recent storm, heavy swimmer load).

Weekly (20–30 minutes): - Test TA and CYA. Adjust TA with baking soda if below 80 ppm. - Inspect pump basket and skimmer basket. Empty both. - Brush the pool walls and steps (stiff bristle brush for plaster/vinyl; softer for fiberglass). - Vacuum the floor if debris is visible. - Check water level — keep it halfway up the skimmer opening. - Inspect pump and motor: listen for unusual noises, check for leaks, verify the pump is priming (water flowing through the clear lid or return jets).

Monthly (30–45 minutes): - Full water test: FC, CC, pH, TA, CYA, calcium hardness. - Deep-clean filter (backwash sand filter; deep-soak cartridge in TSP). - Inspect and clean the pump's hair/lint strainer basket (the one before the pump impeller). - Check the automatic pool cleaner (if applicable) for wear and proper operation. - Lubricate O-rings on pump lid and filter multiport valve with silicone lubricant (not petroleum jelly, which degrades rubber).

Page 8: Seasonal Opening and Closing

Two moments per year make or break your pool's water quality for the entire season.

Spring Opening: Don't just pull off the cover and add chemicals. A proper opening takes 30–60 minutes and sets you up for a clear pool all season.

1. Remove the cover and clean it (hose off debris, let dry, fold and store). A dirty cover stored wet grows algae on the underside, which then goes into your pool.
2. Fill the pool to normal water level if it dropped over winter.
3. Remove winter plugs from returns and skimmer. Reinstall drain plugs in the pump and filter if they were removed.
4. Add a stain-scale prevention product if you have plaster (prevents metal staining from rising calcium).
5. Install the pump strainer basket, ladder grips, and any other equipment removed for winter.
6. Turn on the pump and run on recirculate for 1 hour to aerate and mix.
7. Test water — TA, pH, and CYA are the priorities. Balance them first.
8. Add liquid chlorine to bring FC up to 3–5 ppm.
9. Run the pump 24 hours a day for 48 hours, then retest. If FC is stable and water is clear, switch to normal run times.

10. If the pool is green at opening, proceed with the SLAM method starting with TA balance.

Fall Closing: The goal is to prevent algae growth under the winter cover and make spring opening easy.

1. Run the pump and shock with liquid chlorine to bring FC to 5–10 ppm.
 2. Add a winter algaecide (polyquat 30% or similar; copper-based algaecides can stain and are not recommended for closing).
 3. Lower water level to 6–12 inches below the skimmer.
 4. Add pH buffer if needed to keep TA in range through winter.
 5. Remove, clean, and store ladders, rails, and other metal items.
 6. Drain and store the pump, freeze-proof, with drain plugs removed.
 7. Place the cover (solid covers keep out debris and sunlight; mesh covers allow rain to pass through but require different chemical routines — follow the cover manufacturer's instructions).
 8. Remove, drain, and store return jet fittings and skimmer baskets.
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Page 9: Troubleshooting — When Things Go Wrong

Pool is still green after 7 days of SLAM: Check CYA — if above 80 ppm, you need to drain 30–50% of the pool and refill. High CYA locks out chlorine. No amount of additional shock will clear a pool with locked-out chlorine. Test CYA with the Taylor reagent; do not rely on strips.

Cloudy water after SLAM clears green: Usually means the filter isn't fine enough to capture dead algae particles, or the filter itself is dirty. Try extending run time to 24 hours. Add a filter flocculant (a clarifier that causes tiny particles to clump together so the filter can catch them). If you're using a sand filter, backwash. If cartridge, deep-clean the cartridge with TSP soak.

pH won't stay balanced (bounces up and down): Usually a TA problem. If TA is too low (below 80), pH will be unstable. Raise TA with baking soda first. If TA is correct and pH still drifts, check for new plaster dust or calcium issue. In some cases, aeration from waterfalls, fountains, or return jets pushes pH up — this is normal in pools with water features.

Strong chlorine smell: Most people assume strong chlorine smell means too much chlorine. Usually it means too little — the smell is chloramines, which are the byproduct of chlorine reacting with contaminants (sweat, urine, sunscreen, leaves). If

it smells like a public pool, shock the pool (SLAM-level chlorine) and retest CC. If CC is above 0.5 ppm, the pool needs more chlorine.

Algae keeps coming back: Two possibilities: CYA is too high (drain and refill with fresh water to lower CYA), or circulation is poor in certain areas (add directional returns or brush dead zones more frequently). Persistent algae in corners and steps is almost always a circulation or CYA problem.

Page 10: Next Steps — Your Action Plan

You now have a complete system for clearing your green pool and keeping it clear all season. Here's the sequence:

This week: 1. Buy a Taylor K-2006 liquid test kit if you don't have one. (\$30–\$45. Do not use test strips.) 2. Test your water: TA, pH, CYA, FC, CC. Write down every number. 3. Balance TA to 80–100 ppm using baking soda (1.5 lbs per 10,000 gallons raises TA ~10 ppm). 4. Adjust pH to 7.2–7.6 using muriatic acid (to lower) or soda ash (to raise). 5. If CYA is above 80 ppm: drain 30–50% of the pool before proceeding. 6. Begin SLAM process: add liquid chlorine to reach your target FC based on the Taylor chart and your CYA level. 7. Run the pump 24 hours/day. Set valve to FILTER. 8. Brush the entire pool — walls, floor, steps.

Next 3–7 days: 9. Retest FC each morning and evening. Maintain SLAM-level FC until CC is 0.5 ppm or below and FC holds overnight (less than 1 ppm loss from dark to light). 10. Brush daily. Skim debris daily. 11. Once clear: lower FC back to maintenance level (2–4 ppm depending on CYA) using the chlorine/CYA chart.

Ongoing: 12. Test FC and pH every 2–3 days. Adjust as needed. 13. Full test (FC, pH, TA, CYA) weekly. 14. Monthly: deep-clean filter, inspect equipment, lubricate O-rings. 15. Fall close properly. Spring open properly.

Your Bonus: Included

Your Pool Chemistry Quick Reference Card is included in your Lemon Squeezy library.

This one-page card has every target chemical level, every chemical's dosing rate, and the complete SLAM protocol on a single laminated sheet you can keep by your pool equipment.

Download it free from your Lemon Squeezy library: [Lemon Squeezy — Member's Area]

For new customers purchasing the guide directly: [Get the Full Guide on Lemon Squeezy]

Both files are available in your Lemon Squeezy library. Your Bonus: the companion checklist is included at no extra cost.